

BAAQAR NAQI

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Summary

CS graduate specializing in ML systems (computer vision, NLP, ensemble modeling) and distributed backend engineering (WebRTC, on-device inference, systems programming); seeking SDE, ML engineering, or backend engineering roles.

Education

Jaypee University of Engineering and Technology — *Guna, MP*

Aug 2022 – Jun 2026

B.Tech, Computer Science & Engineering GPA: 9.20 / 10.0

Coursework: Machine Learning, Data Structures & Algorithms, Operating Systems, DBMS, AI Applications

City Montessori School — *Lucknow, UP*

2020 – 2022

ISC Grade 12 (CS elective) — 95.0% ICSE Grade 10 — 98.4% High Distinction, ACER International Benchmark Test (English)

Technical Skills

Languages Python, C, C++, Java, JavaScript, MySQL

ML / AI PyTorch, TensorFlow, scikit-learn, OpenCV, ONNX Runtime, Computer Vision, NLP, CNNs, LSTMs, Transformers

Backend & Web Flask, Django, Node.js, Socket.io, WebRTC, REST APIs

Tools & Systems Git, Linux, Bash, Docker, Web Workers, AudioWorklet

Competitive LeetCode Knight — max rating 1900, 750+ problems solved

Projects

Dyslexia Accessibility ML System — *Research Project* [Code] [Demo] *Python, PyTorch, OpenCV, Flask* 2024/2026

- Engineered an end-to-end ML pipeline to detect dyslexic handwriting in raw photographs and generate plain-language PDF reports for educators without specialist equipment or clinical training.
- Designed a 3-model ensemble — MLP letter classifier (EMNIST, ~800K samples), CNN reversal detector (PyTorch + AMP, 91.86% accuracy), and Bidirectional LSTM for sequence anomaly detection — outperforming baseline OCR by 25% across 500 real-world samples.
- Devised an OpenCV preprocessing pipeline (adaptive Gaussian thresholding, morphological closing) for non-scanned input; uncovered a critical EMNIST orientation bug (90° rotation plus horizontal flip) that produced clean benchmarks while silently breaking real-world inference.
- Formulated a sliding-window reversal clustering algorithm calibrated to clinical prevalence rates (Isa et al., 2019), dynamically weighting local peak density against global reversal rate as sample size grows.
- Containerized with Docker and deployed on Hugging Face Spaces for public, reproducible inference.

PeerSpace — *Serverless P2P Collaboration Platform* [Code] [Demo]

JavaScript, WebRTC, Node.js 2026

- Architected a serverless P2P collaboration suite (video, chat, file transfer, shared document, live transcription) with zero media routed through a server; coded the complete WebRTC signaling stack (SDP offer/answer, ICE negotiation) from scratch using Node.js and Socket.io.
- Structured a full-mesh topology for up to 4 peers across 5 named DataChannels, including a UDP-style unreliable channel for latency-sensitive signals; implemented chunked binary transfer with a custom framing protocol and sender-side backpressure control.
- Provisioned a TURN fallback fetching temporary credentials from an async REST endpoint, so connections traverse symmetric NATs and corporate firewalls while preserving end-to-end DTLS/SRTP encryption.
- Deployed Moonshine ONNX (speech-to-text, 5–15× faster than Whisper in-browser) and DistilBART (meeting summarization) in Web Workers, ensuring no user data leaves the browser; resolved a WebRTC ICE race condition by buffering candidates until `setRemoteDescription()` completed.

Professional Experience

Summer Intern — Honda Cars India Ltd.

Jun 2025 – Jul 2025

- Developed a Python automation script to consolidate recurring departmental data reports, eliminating ~2 hrs/week of manual effort; presented the design and time-saving projections to department stakeholders.
- Audited the department's file-system architecture and delivered a restructuring proposal using the FURPS+ framework (functionality, usability, reliability, performance, supportability, plus cost and constraints); ranked 1st among 6 interns on solution clarity and feasibility.

Leadership

Deputy General Secretary — Development & Innovation Center, JUET

Aug 2025 – Jun 2026

- Directed operations across two technical clubs (RoSPinoT — ML & robotics; VR/AR/MR Club) and spearheaded hackathons, workshops, and programming competitions for 50+ active student members.
- Coordinated project roadmaps across both clubs, mentoring leads and aligning deliverables with event timelines.

Certifications

Machine Learning Specialization — Stanford / Coursera CS50x — Harvard University Introduction to Psychology — Yale / Coursera